Received 19 May 2025
Revised 16 September 2025
1 December 2025
22 December 2025
Accepted 10 January 2026

"I feel more capable": vernacular smartphone affordances among illiterate adults in rural China

Qingting Zhao
School of Journalism and Communication, Shandong University, Jinan, China, and
Hao Gao
School of Journalism and Communication, Nanjing Normal University, Nanjing, China

Abstract

Purpose – Despite significant progress in global literacy campaigns, millions of adults remain illiterate, particularly in rural areas. With the ubiquity of smartphones, illiterate and semi-literate populations are increasingly engaging with digital technologies. This study examines the vernacular affordances these users generate in everyday smartphone use and investigates how such affordances shape their cognitive, emotional, and social experiences.

Design/methodology/approach – The study draws on participatory observation and semi-structured interviews with 17 illiterate and semi-literate individuals in a rural village in Shanxi Province, China. Data were analyzed through thematic analysis guided by the framework of vernacular affordances.

Findings – The analysis identifies three vernacular affordance characteristics—from hidden to perceptible, accidental, and incomplete—and explains how they are realized through three pathways: intergenerational digital support ("digital back-feeding"), symbol–meaning matching, and practice through repetition. These pathways help transform initially opaque smartphone functions into perceptible ones, while also producing affordances that remain unstable or only partially realized. Despite these limitations, smartphone use enhanced informal learning, micro-literacy development, emotional intimacy, and social participation.

Originality/value – The study advances affordance theory by contextualizing vernacular affordances in a rural, low-literacy environment and by demonstrating their unstable, fragmented, and culturally embedded nature. It also shows that the classification of affordances integrates both theoretical insights and empirically derived characteristics. In practice, the findings highlight the importance of intergenerational digital mentorship, culturally sensitive interface design, and policies that promote not only access but also meaningful digital inclusion.

Keywords Illiteracy, Media use, Vernacular affordance, Rural China, Digital inclusion

Paper type Research article

1. Introduction

Illiteracy remains a significant barrier to participation in modern society, as it limits access to information, employment, and social opportunities (Payne, 2006; Fanta-Vagenshtein, 2008). Although global literacy rates have steadily improved, an estimated 754 million adults worldwide remain illiterate, most of them women (UNESCO Institute of Statistics, 2016). The lowest literacy rates are found in South Asia, West Asia, and Sub-Saharan Africa (World Population Review, 2025). In China, the government has made remarkable progress in eradicating illiteracy through successive literacy campaigns since 1949 (Bellard, 1999; Zhou, 2005). According to the Seventh National Population Census, the illiteracy rate fell to 2.67%, with 37.75 million adults still unable to read (Chinese Government Network, 2021). However, these figures may underestimate the actual extent of illiteracy and semi-literacy due to self-reporting and administrative pressure to meet literacy quotas (Lavely *et al.*, 1990; Stites, 1995). Illiteracy remains more prevalent in rural areas, especially among women and ethnic minority groups with limited access to education (Bellard, 1999; Rong and Shi, 2001).

Funding: This work was supported by Annual Project of Guangxi Philosophy and Social Science Research (award number: 24XWC002).

Information Technology & People
Vol. 39 No. 5, 2026
pp. 2266-2284
© Emerald Publishing Limited
e-ISSN: 1758-5813
p-ISSN: 0959-3845
DOI 10.1108/ITP-05-2025-0700

Downloaded from www.emerald.com/itp/article-pdf/39/5/2266/11222077/itp-05-2025-0700en.pdf by University of Guelph user on 04 October 2026

Illiteracy is generally defined as the inability to read, write, and comprehend short passages related to daily life (UNESCO, 1978). Many of those classified as literate are, in fact, semi-literate, struggling to read or write fluently (Islam *et al*., 2023). Although distinct, both groups face similar challenges in digital environments where most tasks presuppose literacy. For analytical consistency, this study collectively refers to them as illiterate groups. With the gradual penetration of technologies and digital terminals, illiterate groups are, in fact, integrating into the digital society in various ways (Hayat *et al*., 2021; Mir and Wani, 2025). Over the past decade, the proliferation of affordable smartphones has expanded ICT access among marginalized populations, fulfilling earlier predictions (Knoche and Huang, 2012). Smartphones are now indispensable across social boundaries of age, gender, and residence (Garg, 2021). Compared with computers, they are cheaper, easier to operate, and more accessible, allowing groups who had never used the Internet via computers to go online for the first time (Correa *et al*., 2020). However, much remains unknown about how these groups actually use digital technologies once they gain access. This puzzle raises the overarching research question of this study:

RQ. How do individuals without fluent literacy manage to navigate and appropriate devices designed for text-based interaction?

To address this puzzle, the concept of vernacular affordance is applied, emphasizing how users perceive and enact the action possibilities that technologies offer in specific contexts (McVeigh-Schultz and Baym, 2015). Examining illiterate groups through this lens enables theorizing how marginalized users transform smartphones into usable tools despite cognitive and structural constraints. Accordingly, the study pursues three analytical aims: (1) identify the vernacular affordances enacted by rural illiterate users; (2) understand how these users, despite lacking formal literacy, navigate and appropriate smartphone functions in everyday life; (3) explain how these vernacular affordances reshape their experiences of learning, emotional support, and social participation. The three analytical aims operationalize the overarching research question.

This study makes three key contributions. Theoretically, it advances the concept of vernacular affordance for understanding marginalized users. Empirically, it contributes evidence from rural China. In practice, it offers insights into inclusive technology and digital policy design.

The remainder of this paper is structured as follows: Section 2 reviews the literature on smartphone use among illiterate populations and Vernacular Affordance. Section 3 specifies the research methods, the selection of the research site, and the data collection and analysis processes. Section 4 presents the findings related to the identification and enactment of vernacular affordances. Section 5 examines the outcomes of these affordances—specifically the cognitive, emotional, and social transformations they generate among illiterate users. The sixth section discusses the findings, and the seventh section concludes the paper.

2. Literature review

2.1 Illiterate populations and new media usage

Digital technologies are typically designed with educated users in mind (Jan *et al*., 2020; Islam *et al*., 2023). If illiterate groups cannot engage with mobile applications, they risk being excluded from the benefits of information and communication technologies (ICTs), thereby deepening the digital divide (Ahmed *et al*., 2019; Islam *et al*., 2023). The advent of smartphones has shifted research on digital inclusion from computer-based solutions to more portable and accessible mobile technologies. To facilitate the digital inclusion of illiterate populations, prior studies have highlighted the role of inclusive design features in smartphones. These include audio prompts, voice input, and icon-driven interfaces that reduce users' dependence on text (Knoche and Huang, 2012; Ahmed *et al*., 2019; Hayat *et al*., 2021; Iqbal, 2021; Doulani *et al*., 2022; Alfayez *et al*., 2023; da Silva *et al*., 2024).

Different interventions have sought to address specific usability challenges. Text-free SMS systems and alternative contact-management strategies enable basic communication tasks for

Downloaded from www.emerald.com/itp/article-pdf/39/5/2266/11222077/itp-05-2025-0700en.pdf by University of Guelph user on 04 October 2026

low-literate users (Hayat *et al*., 2021; Iqbal, 2021). Non-textual websites and local-language interfaces facilitate information retrieval (Doulani *et al*., 2022). Reading-aid applications that combine OCR and translation technologies support access to written content for illiterate and visually impaired users (Alfayez *et al*., 2023). Intelligent voice assistants offer further potential but still struggle with the non-standard vocabulary and grammar often used by illiterate audiences (da Silva *et al*., 2024).

Beyond functional access, mobile phones have also been shown to shape the broader socio-technical practices of illiterate populations. For example, studies of rural illiterate women in India demonstrate how mobile devices expand communication networks and access to critical information (Garg, 2021). Mobile health initiatives, such as MatHealth, illustrate how digital tools bridge gaps between illiterate users and healthcare providers (Musiimenta *et al*., 2021). Recent research on street vendors in Kashmir shows that although mobile payments are widely adopted, illiterate vendors continue to rely on intermediaries and face persistent skill gaps (Mir and Wani, 2025). Collectively, these studies highlight that smartphones not only facilitate technical access but also affect social and economic participation.

Despite these advances, existing scholarship remains concentrated in South Asia and is framed mainly around technological enablement—that is, designing systems to help illiterate populations operate devices. Far less attention has been paid to how these populations themselves navigate, appropriate, and create distinctive digital practices in their everyday socio-cultural contexts. Emerging studies suggest that illiterate individuals demonstrate agency and creativity in their use of smartphones (Chipchase, 2008; Ahmed *et al*., 2013; Mir and Wani, 2025), yet these works only describe isolated cases rather than systematically analyzing the mechanisms through which such practices emerge. Thus, the literature provides only a partial understanding of illiterate users' digital engagement. Addressing this gap is important because existing definitions of digital inclusion remain incomplete without accounting for the culturally grounded, practice-based strategies through which illiterate users make technology meaningful.

2.2 Vernacular affordance

To address the gap identified above, this study turns to vernacular affordances as a framework for understanding user-driven, culturally situated practices. The concept of affordance was first introduced by Gibson (1979) to describe what the environment offers to an organism, whether beneficial or harmful. Later, Norman (2013) emphasized perceived and actual properties of objects that determine how they can be used, making the concept influential in design studies and Human-Computer Interaction (Bucher and Helmond, 2018). Over time, distinct streams of affordance research have emerged. Imagined affordances emphasize the material, mediated, and emotional dimensions of technology (Nagy and Neff, 2015), while communicative affordances highlight portability, availability, and multimodality as key factors shaping interaction (Schrock, 2015). These perspectives collectively illustrate that affordances are neither technologically determined nor entirely socially constructed, but rather emerge at the intersection of material features and user practices.

Building on this trajectory, McVeigh-Schultz and Baym (2015) introduced the notion of vernacular affordance to capture how technologies are appropriated through users' everyday practices and meaning-making processes, rather than being predetermined by designers. Affordances, from this perspective, shift from "hidden" to "perceptible" through repeated engagement, as users discover latent possibilities or develop new practices beyond design intentions. Instead of assigning affordances externally, this approach emphasizes how users themselves articulate the action possibilities of technologies in culturally specific ways (Bucher and Helmond, 2018). This perspective highlights user agency in articulating action possibilities, yet such agency is not unconstrained. Among illiterate users, the realization of vernacular affordances is contingent upon their literacy-related capabilities—such as symbolic recognition, memory-based learning, and adaptive strategies—which determine whether hidden affordances become perceptible or remain accidental and incomplete.

Downloaded from www.emerald.com/itp/article-pdf/39/5/2266/11222077/itp-05-2025-0700en.pdf by University of Guelph user on 04 October 2026

Empirical research has demonstrated the contextual nature of vernacular affordances. For example, Darling-Wolf (2021) showed how low-income groups in a rural Japanese community used digital media to maintain family ties and imagine global connections, while also experiencing alienation from urban populations perceived as more technologically adept. These findings highlight that affordances are locally situated and culturally contingent. By extension, illiterate populations in rural China are also likely to develop distinctive practices and perceptions in their smartphone use, producing vernacular affordances that reflect their own socio-cultural environment.

2.3 Theoretical premises

Taken together, the reviewed literature highlights both the opportunities and limitations of current approaches. Most prior work on illiteracy and digital technologies has emphasized technical interventions to reduce literacy demands. While valuable, this perspective provides only a partial understanding of digital inclusion. Affordance theory broadens the scope by considering not just functional access but also the emergent, user-driven practices that arise in specific contexts. Within this body of work, vernacular affordance offers a particularly appropriate lens, foregrounding the situated strategies through which marginalized groups interpret and repurpose technologies.

This study adopts the concept of vernacular affordance to examine how illiterate populations in rural China engage with smartphones. It is argued that these groups generate practices that produce hidden affordances — perceptible, accidental, or incomplete. Such practices influence not only technology use but also broaden aspects of cognitive development, literacy learning, and social participation. The conceptual model underpinning this study is presented in Figure 1.

The classification of vernacular affordances used in this study draws on both theoretical and empirical foundations. The idea of affordances shifting from “hidden” to “perceptible” is informed by McVeigh-Schultz and Baym’s (2015) discussion of how latent possibilities become noticeable through extended user engagement. Our findings reflect this process among rural illiterate users as they gradually discover unnoticed smartphone functions. In contrast, the categories “accidental” and “incomplete” affordances emerged solely from the empirical data. These concepts describe affordances that users encounter unintentionally or can only partially operationalize due to literacy-related barriers. Thus, the three categories together capture both theoretically grounded and data-driven characteristics of vernacular affordances.

Building on the three analytical aims outlined earlier—identifying vernacular affordances, understanding how they enable navigation and appropriation, and explaining their broader experiential outcomes—the conceptual model in Figure 1 traces a sequential process from socio-cultural context, through users’ vernacular practices, to the characteristics of vernacular affordances and their outcomes. The author develops a model to operationalize vernacular affordances among rural illiterate smartphone users.

3. Methods

This study adopts a qualitative research design to gain an in-depth understanding of the lived experiences and situated practices of illiterate smartphone users. This approach is particularly



Figure 1. Conceptual model of vernacular affordances in illiterate smartphone use. Source: Authors’ own work

Downloaded from www.emerald.com/itp/article-pdf/39/5/2266/11222077/itp-05-2025-0700en.pdf by University of Guelph user on 04 October 2026

suitable for exploring the nuanced ways in which users navigate and make sense of technology in their everyday lives, a phenomenon that quantitative methods would struggle to capture.

3.1 Selection of the research site

This study primarily employed participatory observation and semi-structured interviews. Fieldwork was conducted in Village S, Shanxi Province, China, from December 15, 2022, to February 10, 2023. Village S consists of 447 households and 1,277 residents. Its economy is relatively well developed, primarily based on agriculture, but it is gradually transitioning to urbanization, with an internet penetration rate of 90%. Village S was chosen as the research site because it represents a common phenomenon in rural Shanxi: a relatively high level of digital penetration alongside the continued presence of illiteracy. Although not statistically representative, this unique environment provides a solid foundation for examining how illiterate and semi-literate populations engage with smartphones. According to the Seventh National Population Census (National Bureau of Statistics of China, 2021), illiteracy rates across northern provinces are relatively similar, suggesting that the case of Village S reflects broader regional trends.

3.2 Population and sample

Recruiting participants posed a significant challenge, as illiterate individuals are often reluctant to disclose their literacy status and can be difficult to identify through formal channels. To address this, snowball sampling was adopted, as it is particularly suitable for hidden populations. By relying on trusted intermediaries within the community, this approach facilitated access, built rapport, and minimized potential stigmatization. In total, 17 participants were interviewed (nine illiterate and eight semi-literate; 13 women and four men; see Table 1). The predominance of women reflects rural demographic patterns, as women account for a larger share of the illiterate population. National data confirm this gender gap: in Shanxi Province, the illiteracy rate among women aged 15 and above is 2.39%, compared to 0.81% among men (National Bureau of Statistics, 2024). The age distribution of the sample is similarly skewed toward older adults. This pattern reflects a broader demographic reality: in rural China, illiteracy is concentrated almost entirely among the older generation due to historical limitations in educational access. Following the founding of the People's Republic of China, multiple nationwide literacy campaigns were launched, and overall educational levels increased substantially, making younger generations almost entirely literate. Consequently, the observed age bias is not an artifact of the recruitment process, but rather an inevitable outcome of structural educational conditions in rural China.

3.3 Data collection

In addition to semi-structured interviews, participatory observation was conducted by spending extended time in participants' homes and community environments. Researchers observed how participants used smartphones in everyday activities, such as making video calls, browsing short-video apps, and attempting online shopping. These observations complemented interview data by capturing practices that participants might not have explicitly articulated, offering a richer understanding of how affordances were enacted *in situ*. Interviews lasted 20–40 min and were conducted in the local dialect. All interviews were audio-recorded with participants' informed consent. To ensure anonymity and privacy, personal identifiers were removed during transcription, and each participant was assigned an alias (S1–S17). Audio files, transcripts, and field notes were stored on encrypted drives accessible only to the research team. No raw recordings or identifiable data were shared outside the research group. These measures ensured that participants' information was handled securely and confidentially throughout the research process. Interviews lasted 20–40 min and were conducted in the local dialect. All interviews were audio-recorded with participants' informed

Downloaded from www.emerald.com/itp/article-pdf/39/5/2266/11222077/itp-05-2025-0700en.pdf by University of Guelph user on 04 October 2026

Table 1. Demographic information of respondents

Respondent	Age	Gender	Ethnicity	Occupation	Education level	Reason for low education level
S1	70	Male	Han	Farmer	Grade 1	Poverty, dislike for school
S2	53	Female	Han	Housewife	Grade 2	Poverty, poor academic performance
S3	53	Male	Han	Tire factory worker	Grade 2	Dropped out to learn opera singing
S4	73	Male	Han	Farmer	Grade 1	Personal unwillingness to attend school
S5	72	Female	Han	Housewife	No formal education	Poverty took care of the younger brother
S6	65	Female	Han	Housewife	No formal education	Personal unwillingness to attend school
S7	71	Female	Han	Housewife	Grade 1	Personal unwillingness to attend school
S8	50	Female	Yao	Part-time at the distillery	No formal education	Parents did not allow schooling
S9	51	Female	Yao	Self-employed	No formal education	War outbreak
S10	52	Female	Yao	Housewife	No formal education	Poverty, parents did not allow schooling
S11	54	Female	Han	Housewife	Grade 2	Personal unwillingness to attend school
S12	35	Female	Yao	Housewife	No formal education	Took care of older adults at home
S13	58	Female	Han	Housewife	No formal education	For many children, parents did not allow schooling
S14	52	Male	Han	Pig farmer	Grade 2	Personal unwillingness to attend school
S15	75	Female	Han	Housewife	No formal education	Parents did not allow schooling
S16	70	Female	Han	Housewife	Grade 2	Poverty took care of family members
S17	73	Female	Han	Housewife	No formal education	Poverty, parents did not allow schooling

Source(s): Authors' own work

consent. Data collection concluded after 17 participants, as thematic saturation was reached between the fifteenth and seventeenth interviews, indicating that no substantially new insights emerged (Guest *et al*., 2006).

3.4 Data analysis

All recordings were transcribed within one week. The dialect content was translated into Mandarin, and accuracy was ensured through cross-checking by multiple researchers.

This study employed thematic analysis (Braun and Clarke, 2006), which was considered appropriate due to its flexibility in integrating both inductive insights and deductive links to theory. Although the vernacular affordance framework guided the study, open coding was used initially to avoid imposing pre-defined categories and to allow novel, context-specific practices to emerge from the data. This approach balances data-driven discovery with theory-informed interpretation, consistent with prior affordance-based studies (Thapa and Sein, 2018; Hazra and Priyo, 2022).

Two researchers independently coded all transcripts and observation notes, generating an initial set of descriptive codes. These codes were compared, merged, and refined through

Downloaded from www.emerald.com/itp/article-pdf/39/5/2266/11222077/itp-05-2025-0700en.pdf by University of Guelph user on 04 October 2026

iterative discussion. At this stage, the codes were examined against the vernacular affordance framework, not as a rigid template but as a sensitizing concept (Blumer, 1954) to interpret how users' sense-making strategies aligned with or extended theoretical categories. Themes and sub-themes were distilled through repeated comparison and negotiation, with reciprocal audits between coders to ensure reliability and credibility. For example, the excerpt "I can watch TV dramas on Kuaishou now. I don't know how I found it . . . it just came up when I was swiping" (S1) was first coded as "algorithm dependence" and "unable to search". These initial codes were then grouped into a more focused code capturing "accidental discovery," which in turn contributed to the higher-level theme of "accidental affordances." This analytic trajectory exemplifies how instances in the data were progressively abstracted and interpreted through the lens of vernacular affordances. Additional coded examples that follow a similar analytic logic are presented in Appendix 1.

This iterative process produced core themes illustrating how illiterate and semi-literate individuals navigate smartphone use and how their situated practices give rise to distinctive forms of vernacular affordance. Analytical saturation was reached when no substantially new categories emerged.

4. Findings

The interview results indicate that the illiterate population exhibits consistent preferences in their smartphone usage, with the predominant usage pattern being "WeChat + short video platforms (Kuaishou, Douyin)." Additionally, S3, S8, S9, and S12 were able to shop on Pinduoduo and through Douyin live streams, S11 used the Karaoke app "Quanmin K Ge" for online singing, and S8, S9, S13, S14, and S17 utilized Douyin and Kuaishou short video apps for online self-presentation. These popular platforms enable diverse activities such as video creation and social interaction, making them integral to participants' digital practices.

Consistent with the theoretical lens of vernacular affordances, participants' engagement with digital technologies generated three distinct forms of action possibilities: hidden–perceptible, accidental, and incomplete affordances. To align with naming conventions in affordance research (Thapa and Sein, 2018), this study labels the corresponding abilities as perceptibility, discoverability, and partial performability**,** each capturing the specific ways illiterate users realize, encounter, or partially enact these affordances. As summarized in Table 2, these affordances represent the core vernacular action potentials enacted by rural illiterate users, directly addressing Aim a. The analyses throughout Section 4 explain how

Table 2. Vernacular affordances and corresponding ability terms identified among illiterate smartphone users in rural China

Affordance category	Ability term	Definition
Hidden–Perceptible	Perceptibility	The ability to transform initially hidden or opaque technological possibilities into perceptible and actionable functions through imitation, memory-based learning, external assistance, or sustained practice
Accidental	Discoverability	The ability to encounter technological functions unintentionally through coincidence, algorithmic exposure, or trial-and-error, without being able to reproduce the pathway reliably
Incomplete	Partial performability	The ability to perform only part of a digital task when literacy-dependent or cognitively demanding steps remain inaccessible, resulting in fragmented or interrupted digital practices

Source(s): Authors' own work

Downloaded from www.emerald.com/itp/article-pdf/39/5/2266/11222077/itp-05-2025-0700en.pdf by University of Guelph user on 04 October 2026

these affordances are realized through users' situated strategies and everyday practices, thereby addressing Aim b.

4.1 Hidden–perceptible affordances: perceptibility

In this study, perceptibility refers to illiterate users' capacity to transform initially hidden or opaque technological possibilities into recognizable and actionable functions. For illiterate populations, the functionality of a smartphone is not self-evident; many text-based applications and interface features are initially "hidden" because they rely on written labels or symbolic cues that are difficult to interpret. However, through a combination of external support, imitation, memory-based learning, and sustained practice, these hidden possibilities gradually become perceptible and usable. Recognizing this ability helps clarify both what hidden–perceptible affordances are present and how they are realized through everyday interaction. This study identified three key mechanisms that facilitate this transition: digital back-feeding, symbol–meaning matching, and practice through repetition. Together, these mechanisms explain how illiterate users make invisible affordances visible in their daily digital routines.

The first is digital back-feeding, in which support from younger generations makes technology more accessible (Zhou and Ding, 2020). Most participants, aged between 50 and 75, relied on their children for both new and second-hand smartphones. This act, rooted in the Confucian value of filial piety, is often the starting point for their digital journey. Children primarily give their parents smartphones to strengthen their connection. As China's urbanization advances, many young people leave their hometowns to work in cities, leaving older adults as the primary residents in rural areas. Separation between elderly parents and their children has become the norm, a pattern reflected in the village survey as well. Maintaining contact with their children becomes an essential yet challenging task for older adults, particularly for illiterate individuals. Illiterate people, excluded from the world of written communication, were previously unable to stay in touch with family and friends through letters or text messages. WeChat video calls provided a new way for them to connect with their families, eliminating the illiteracy-related barriers to writing and reducing concerns about high phone bills. As respondent S5 explained, this new form of communication provides reassurance and enhances emotional bonds:

> My smartphone is a second-hand phone that my children gave me. The kids are not around me, but they want me to have it so we can video call . . . In the past, we could only make phone calls, couldn't see each other, and had to keep the phone bill in mind so we wouldn't say much. Video calls are much more convenient. (S5)

The second mechanism is a cognitive strategy of matching symbols with meanings. Illiterate users learn to identify app functions using graphical symbols, a crucial method for navigating interfaces. For instance, many users prefer to set their photos as their WeChat profile pictures. Illiterate users who cannot read WeChat nicknames rely on profile pictures to identify the people they want to contact. The WeChat profile picture, therefore, effectively replaces the traditional "phonebook" or contact list, which typically requires literacy, providing a unique way for illiterate individuals to identify their contacts. However, if infrequently contacted individuals do not use their photos as profile pictures, it becomes challenging for illiterate users to recognize those contacts. As S9 noted, "*For people I contact frequently, I know who they are by looking at their profile pictures. For those I don't contact often, I can't recognize them*."

The third and most crucial mechanism is practice through repetition. Users master core functions by performing the same actions daily, such as making WeChat calls or watching short videos. Through prolonged repetition, they develop a form of muscle memory, sustaining their access to the smartphone's affordances. It was demonstrated by S2, who could recognize the characters for frequently used functions like "red packet" but not for others like "video

Downloaded from www.emerald.com/itp/article-pdf/39/5/2266/11222077/itp-05-2025-0700en.pdf by University of Guelph user on 04 October 2026

call," explaining, "I remember these characters now because others taught me, and I remember what they are used for, but I can't write them down—it's all from memory." Interestingly, except for S7 and S10, the other respondents mentioned that they believed their memory was better than others. Stauffer *et al.* (1978) noted that anecdotal evidence suggests that illiterate adults in tribal societies possess a highly developed ability to recall verbal information. The interview results of this study similarly indicated that the confidence of illiterate individuals in their memory played an essential role in mastering smartphone use, particularly in the repeated memorization of icons and function locations. This memory ability helps overcome literacy barriers to some extent.

4.2 Accidental affordances: discoverability

For illiterate users, even after mastering basic smartphone operations, many actions emerge not through deliberate search but through coincidence, trial-and-error, or algorithmic exposure. This phenomenon is captured by what we refer to as *discoverability*, which corresponds to accidental affordances, referring to features that users encounter unintentionally and cannot reliably reproduce. When navigating short video platforms, for instance, participants often swipe aimlessly, passively consuming content recommended by the algorithm. As several interviewees remarked, "I don't know how it happened," indicating that they might stumble upon an interface or function once but have no way to find it again. Introducing discoverability helps clarify the contingent, serendipitous nature of these encounters and explains how accidental affordances arise in their everyday digital routines.

S1 is a 70-year-old man who attended only one school year and made a living as a farmer. He lives with his hard-of-hearing wife, and his son, who is rarely at home, purchased his smartphone. S1 learned how to use the phone from an older villager who was more proficient with smartphones. During the interview, S1 mentioned that he could use Kuaishou to watch movies. The researchers then asked him to demonstrate how he operated the phone. He opened the Kuaishou app but could only swipe up and down, not knowing how to search because he couldn't read. He could not find the movie interface despite swiping for an extended period. It was evident that watching movies on the short video platform was purely accidental, relying primarily on "luck." Consequently, most of the time, he could only passively consume content recommended by the app, without being able to search for videos he wanted to watch actively. It highlights a key aspect of their digital experience: moments of discovery are often fleeting. They cannot be converted into stable, repeatable skills without the textual cues that literate users rely on.

Similarly, S13, the only participant to have hosted a livestream, could not reliably recall the steps to do so. S13 is a 57-year-old woman from a family that values males over females. The family has nine children, but only her brothers were allowed to attend school. She has never received a formal education and cannot even write her name. Among the respondents, she is very proficient in using smartphones and is the only one to have hosted a livestream. She mentioned that her purpose for live streaming was to chat with people she knows, who would join when they saw her go live. The researchers then asked her to demonstrate how to start a livestream, but she momentarily forgot the specific steps. Because she could not read, she would forget the procedure if she did not use it regularly. The researchers taught her how to start a livestream again, and shortly after that, she went live. Ten minutes later, her sister joined the livestream. After briefly inquiring about the researchers' visit, the sisters began chatting.

In contrast, the livestream provided a fleeting moment of connection with her family; her inability to remember the text-based procedure made such online gatherings accidental and unsustainable. Her experience demonstrates the fragility of these skills. At the same time, she possessed the social desire and initial capability; the lack of a reproducible, text-free pathway meant the affordance for self-presentation remained precarious. Ultimately, the accidental nature of these affordances confines users to a digital environment characterized by

Downloaded from www.emerald.com/itp/article-pdf/39/5/2266/11222077/itp-05-2025-0700en.pdf by University of Guelph user on 04 October 2026

serendipity rather than control, limiting their ability to leverage the technology's potential fully.

4.3 Incomplete affordances: partial performability

For illiterate users, many action possibilities remain only partially realizable. This phenomenon is captured by what we refer to as partial performability, which corresponds to incomplete affordances. These affordances arise when users can perform some steps of a digital task but are unable to complete more text-dependent or cognitively demanding stages. As a result, their digital practices often appear fragmented: they can initiate specific actions but cannot extend or finalize them. For example, several respondents could shop online, yet none could independently track their orders because they could not read, requiring them to rely on others for assistance. This pattern illustrates a common scenario in which the first half of a task is achievable, while the second half—typically involving reading or textual navigation—remains inaccessible. Identifying partial performability thus helps explain how illiterate users participate in digital environments while still experiencing structural constraints that limit the depth of their engagement.

S09's experience with Douyin further illustrates this incompleteness. S09 is a 51-year-old housewife from Yunnan Province, China, belonging to the Yao ethnic minority, one of China's minority groups. Her hometown is located on the border between China and Vietnam, and she missed out on schooling during childhood due to the outbreak of war in the region. At 23, she married Village S, where she had a son and a daughter. She frequently uses the "shoot same style" feature to create videos by clicking the interface, even though she cannot read the function's name. While she understands how to create these specific videos, her content options are limited because her illiteracy prevents her from searching for and selecting her favorite songs. It creates a paradoxical situation in which she can participate in a trend but cannot personalize her creation, showcasing a functional yet ultimately constrained form of digital expression. Furthermore, while she understands the comment section's interaction mechanism, her fear of making mistakes with text limits her to using only emojis. She stated:

> I can comment on Douyin but can't type comments, so I just send a heart emoji or a flower. I can't read characters, and I'm afraid of making mistakes—if I make a mistake, everyone can see it in the comment section . . . so I don't dare to leave random comments.

Her strategy of using only emojis is a poignant example of an incomplete affordance in action. It is a clever workaround that allows her to remain socially present and avoid the potential shame of making a public mistake. However, it simultaneously silences her voice, preventing nuanced conversation and reinforcing her marginalization in text-centric online communities. Therefore, incomplete affordances represent a double-edged sword: they provide a crucial foothold for participation in the digital world, but their fragmented nature can also circumscribe the depth of that participation, creating new, subtle forms of digital inequality.

5. Outcomes of enacting vernacular affordances

The enactment of hidden–perceptible, accidental, and incomplete affordances has transformed the lifestyles of illiterate individuals, leading to significant cognitive, emotional, and social changes. These findings primarily address aim c, which concerns how these vernacular affordances reshape their experiences of learning, emotional support, and social participation.

5.1 Cognitive and micro-literacy shifts through informal learning

Illiterate individuals may not have received a formal education or may have had only brief schooling. Smartphones have become new channels for informal learning, enhancing users' literacy skills in subtle ways. S09 reported learning new characters from literacy tutorials on Douyin, while S11 recognized some characters through the karaoke app Quanmin K Ge. Users

Downloaded from www.emerald.com/itp/article-pdf/39/5/2266/11222077/itp-05-2025-0700en.pdf by University of Guelph user on 04 October 2026

also developed creative "weapons of the weak" to overcome literacy barriers. For instance, S14 draws characters he does not recognize in WeChat, sends them to his daughter, and then uses the pronunciation feature to learn how to read them. Only after S14's explanation did the researchers learn about WeChat's "Listen to Text Message" feature. Simply tapping a text message in a chat plays it aloud. Although the researchers themselves are frequent WeChat users, they were unaware of this feature, as literate individuals rarely consider using it in their daily contexts. In contrast, illiterate individuals rely on listening to text, leading them to discover features in WeChat that others may overlook. Such practices enriched vocabulary, practical skills, and confidence. As S11 reflected, illiteracy made her feel "pitiful" and looked down upon, but smartphone use helped her feel "more capable than before."

5.2 Emotional affordances and intimacy

Beyond cognitive benefits, smartphones provided vital emotional support. For participants left behind in rural villages while their children migrated to cities, video calls alleviated loneliness and reinforced intergenerational intimacy. On one hand, they use video calls as a substitute for written communication, allowing them to maintain contact with distant family members and friends. On the other hand, smartphones have transformed their entertainment, helping them temporarily forget their worries and alleviate loneliness caused by living alone or by the pressures of daily life through short videos and access to the online world. S10, who has a leg disability, seldom goes out, but now she can see her family through video calls. She stated: "*My children are all in other cities. Whenever I miss them, I just make a video call. Even though it's not the same as being in person, it still brings some comfort.*" (S10)

S15 is a 75-year-old woman whose children do not live in the village, leaving her to live alone most of the time. Three years ago, her husband passed away, and since then, she has often felt lonely living by herself. She relies on her phone to ease her loneliness when she's alone. Sometimes, she calls her children via video, but she often watches short videos. She mentioned that she has two smartphones, and when she cannot sleep at night, she switches between them to watch short videos. She particularly enjoys videos featuring a young monk who frequently says, "Be in a good mood and respect others." Each time she watches these videos, she feels better.

5.3 Enhanced social participation and spatial confidence

Smartphones expanded users' capacity to participate in community and public life. Video calls reduced the fear of entering unfamiliar environments by enabling real-time navigational support. It broadened their conversational repertoire and strengthened their sense of integration into broader society. As S09 explained, "It's *much more convenient now with video calls. Before, I didn't dare go out because I couldn't read. Now, I dare to go out. When I go out, I start a video call and share it so others know where I am and can tell me how to get there. Before, I could only listen to the person over the phone but couldn't see my surroundings—now, with video calls, I can see right away. In the past, I could ask others, but I was afraid of being deceived, so I wasn't very willing to ask strangers.*" (S09).

Additionally, illiterate individuals can proactively learn about local and distant news through short videos. Before smartphones, people primarily obtained news through word of mouth; now they can access a wide range of news content independently. During the interviews, respondents mentioned that they learned about "what is happening in nearby villages" and news such as "Japan releasing nuclear wastewater into the ocean" through short videos. Some also mentioned the "war between Russia and Ukraine." Short videos have connected them to the outside world, enabling them to use news they encounter as conversation topics and to express their views on specific events, thereby promoting interaction with literate individuals and enhancing social participation.

Taken together, while vernacular affordances enabled illiterate users to gain new forms of learning, intimacy, and participation, their accidental and incomplete nature also

Downloaded from www.emerald.com/itp/article-pdf/39/5/2266/11222077/itp-05-2025-0700en.pdf by University of Guelph user on 04 October 2026

circumscribed the depth and sustainability of these practices, generating subtle yet persistent forms of digital inequality.

6. Discussion and conclusion

6.1 Discussion

This study investigated the smartphone practices of illiterate and semi-literate populations in rural China, aiming to reveal their unique process of generating vernacular affordances. The findings show that illiterate users, through a series of vernacular strategies, transform initially hidden smartphone functionalities into perceptible ones, although these affordances are often accidental and incomplete. This section situates the findings within the broader literature, elaborates on theoretical contributions, and outlines practical implications, limitations, and directions for future research.

Research on illiterate users has repeatedly shown that mobile phones are both essential and challenging (Garg, 2021; Hayat *et al*., 2021; Mir and Wani, 2025). A study of rural Indian women found that although 85% used mobile phones, most did not own one (Garg, 2021). By contrast, all participants in this study owned a personal smartphone, indicating that inequality today lies less in access and more in modes of use. It aligns with second-level digital divide scholarship, which emphasizes skill gaps and differentiated appropriation rather than access alone.

The strategies documented here—identifying functions through graphic symbols and repetitive practice—echo previous findings that illiterate users rely on icons and character shapes (Knoche and Huang, 2012; Medhi *et al*., 2006). Yet, comparative work shows differences: in Pakistan, illiterate users memorized phone numbers to manage contacts (Hayat *et al*., 2021), while in rural China, profile pictures and icons served as substitutes for text. These differences underscore the cultural embeddedness of affordances and highlight the need for comparative approaches.

The role of family support further distinguishes this study. In line with findings from South Asia, where families facilitate mobile payments for illiterate vendors (Mir and Wani, 2025), this study identifies digital back-feeding—where children provide devices and teach parents—as a crucial mechanism. Such support not only lowers entry barriers but also creates a moralized pathway rooted in Confucian filial piety, showing that technological appropriation cannot be separated from cultural values and kinship obligations.

This study finds that smartphone use among illiterate populations yields significant positive outcomes, including acquiring new knowledge, receiving emotional support, and promoting social participation. These findings align with other research highlighting the benefits of smartphone access for illiterate groups, including facilitated interaction with the outside world for illiterate women (Garg, 2021), motivation for literacy development (Doulani *et al*., 2022), enhanced self-confidence and social participation for middle-aged illiterate women (Irizarry *et al*., 2002; Gou and Gong, 2025), and increased productivity for low-literacy users (Iqbal, 2021). It underscores the critical importance of advancing digital inclusion for illiterate populations, as it not only benefits the group itself but also contributes more broadly to promoting digital inclusion and social equity.

6.2 Theoretical contributions

This study makes two main theoretical contributions. First, rather than claiming to propose entirely new types of affordances, this study refines and extends the affordance perspective through a contextualized lens. Prior literature has conceptualized affordances as "imagined" (Nagy and Neff, 2015), "communicative" (Schrock, 2015), and "vernacular" (McVeigh-Schultz and Baym, 2015). Building on this theoretical lineage, this study identifies three affordance characteristics among illiterate smartphone users: hidden-to-perceptible, accidental, and incomplete. These three characteristics have distinct origins: "hidden-to-

Downloaded from www.emerald.com/itp/article-pdf/39/5/2266/11222077/itp-05-2025-0700en.pdf by University of Guelph user on 04 October 2026

perceptible" draws from McVeigh-Schultz and Baym's (2015) discussion on how latent possibilities are gradually perceived by users through sustained interaction; whereas "accidental" and "incomplete" emerged entirely from the empirical data of this study, serving to encapsulate the action possibilities that illiterate users encounter unintentionally or can only partially realize during use.

To align with naming conventions in affordance research (Thapa and Sein, 2018), this study further maps these three characteristics to three corresponding abilities: perceptibility, discoverability, and partial performability. These ability terms do not propose new affordance types; instead, they clarify how illiterate users realize, stumble upon, or partially use these affordances in practice. Taken together, this framework demonstrates that in the context of rural illiteracy, affordances may appear unstable, fragmented, and contingent, thereby offering a nuanced, situated extension to vernacular affordance theory.

Second, this study contributes a "culturally embedded" perspective on affordances. While prior research acknowledges that affordances arise from interaction within specific contexts, it rarely examines how cultural values, kinship structures, and moral obligations shape their emergence. By demonstrating how filial piety, reliance on visual symbols, and memory-based learning strategies jointly influence smartphone use, this study indicates that affordances stem not only from device–user interaction but are also mediated by broader socio-cultural logics. Such a cultural theoretical positioning further deepens vernacular affordance theory, demonstrating that affordances cannot be understood independently of the cultural contexts in which they are embedded.

6.3 Practical implications

The findings yield several important practical implications. First, policy and social interventions should recognize the critical role of digital back-feeding in facilitating smartphone use among illiterate populations. Interventions could be designed to support intergenerational learning networks by formalizing "family digital mentorship" programs in rural areas, encouraging and guiding younger generations to teach their parents or grandparents how to use smartphones. Such programs would build on existing cultural norms of filial piety and strengthen digital inclusion through family-based support structures.

Second, design and technology development should pay greater attention to how illiterate users engage with digital interfaces. Designers need to create flexible, multimodal systems that align with non-standard usage strategies, such as reliance on icons, pictorial cues, and memory. Furthermore, speech-to-text and "listen-to-text" functions should be optimized for local dialects and irregular speech patterns to ensure that illiterate populations are not excluded from the communicative affordances of mobile technologies.

Finally, digital inclusion programs should move beyond access and emphasize meaningful use. The demonstrated benefits of smartphone use in this study—literacy gains, increased confidence, and greater social participation—highlight the importance of government and NGO initiatives that provide not only devices and connectivity but also opportunities for sustained and empowering engagement. By addressing the second-level digital divide, such programs can ensure that marginalized populations fully benefit from digital transformation.

These insights highlight that digital inclusion policies worldwide must go beyond device provision to ensure that marginalized groups can generate stable, sustainable, and culturally resonant forms of technology use.

6.4 Limitations and future directions

Several limitations must be acknowledged. First, the study was based on a small sample of 17 participants in a single village, limiting generalizability. Second, the sample is skewed toward women and older adults, reflecting the broader demographic reality that illiteracy in rural China is concentrated among these groups due to historical educational constraints. Although this structural bias may shape certain digital practices, it does not alter the study's central

Downloaded from www.emerald.com/itp/article-pdf/39/5/2266/11222077/itp-05-2025-0700en.pdf by University of Guelph user on 04 October 2026

conclusions, which focus on literacy-related constraints and sense-making processes. However, it does limit the extent to which age- or gender-based comparisons can be explored. Third, the qualitative design, reliant on interviews and observations, lacks longitudinal depth. Finally, although this study proposes new dimensions of vernacular affordances, these remain provisional and require further validation in different socio-cultural and technological settings. Future research could address these gaps in several ways. Comparative studies across regions (e.g. South Asia, Sub-Saharan Africa) would test the cross-cultural robustness of the identified affordance characteristics. Longitudinal studies could trace how the practices of illiterate populations evolve, especially regarding literacy acquisition and intergenerational dynamics. Further, experimental and design-based studies could assess how interface modifications (e.g. symbol-based navigation) affect usability and empowerment among illiterate groups.

7. Conclusion

In summary, this study demonstrates how illiterate populations in rural China creatively negotiate smartphone use through localized practices, generating vernacular affordances that range from hidden to perceptible, accidental, and incomplete. These findings are further articulated through the corresponding user abilities—perceptibility, discoverability, and partial performability—which capture how illiterate users recognize, encounter, or partially enact these action possibilities in everyday digital life. By situating these findings within the broader affordance framework, the study both localizes and destabilizes the concept of vernacular affordances, showing their fragmented, contingent, and culturally embedded nature. Practically, the study underscores the urgency of designing inclusive technologies and policies that recognize the agency of illiterate users while supporting intergenerational learning. Theoretically, it enriches affordance scholarship by foregrounding instability, partiality, and cultural mediation. Together, these contributions highlight the importance of ensuring that marginalized populations worldwide can not only access but also meaningfully participate in the digital age.

Acknowledgments

The authors would like to express their sincere gratitude to all the participants in this study for their time and cooperation.

(The Appendix follows overleaf)

Downloaded from www.emerald.com/itp/article-pdf/39/5/2266/11222077/itp-05-2025-0700en.pdf by University of Guelph user on 04 October 2026

Table A1. Illustrative coding table

Theme	Sub-theme	Data excerpt	Initial code	Focused code
Hidden–Perceptible Affordances	Digital Back-feeding and Technological Introduction	"My children gave me the old smartphone they replaced . . . and taught me how to make video calls." (S5)	Children giving phone; Children teaching	Digital back-feeding; Intergenerational tech guidance
		"I didn't know how to use it, everyone taught me—whoever knew, I asked them." (S2)	Multi-source teaching; Community help	Village mutual-aid learning
		"My children worry I'll be lonely, so they want me to use the phone to watch videos." (S5)	Emotional motivation; Family expectations	Family-driven digital engagement
		"My kids taught me how to scan QR codes many times, but I still can't remember." (S10)	Difficulty in cross-generational learning	Intergenerational tech transfer difficulty
	Symbol–Meaning Matching	"I recognize people by their profile pictures; I can't read their names." (S9)	Recognizing contacts by photos	Image-as-text substitution
		"I don't know the Chinese character for the 'plus' icon, but I remember tapping it to record videos." (S10)	Remembering icon position	Spatial-positioning strategy
		"I know what the functions are by looking at the icons." (S2)	Icon recognition	Image-based navigation
		"I can't recognize the profile pictures of people I don't often contact." (S5)	Recognition limitations	Limits of image-based identification
	Repeated Practice and Procedural Mastery	"I can't write the characters for red packet, but I remember where to tap." (S2)	Position memory; Repeated action	Procedural action memory
		"I tap on videos every day, so I know how to do it." (S14)	Habitual behavior	Repetition-based mastery
		"I remember it after tapping many times." (S8)	Reinforced memory	Learning by repetition
		"I swipe up on Kuaishou every day, so I know how to scroll." (S3)	Fixed usage pattern	Stable operation routine

(*continued*)

Downloaded from www.emerald.com/itp/article-pdf/39/5/2266/11222077/itp-05-2025-0700en.pdf by University of Guelph user on 04 October 2026

Table A1. Continued

Theme	Sub-theme	Data excerpt	Initial code	Focused code
Accidental Affordances	Algorithm-driven Discovery	"I see his videos every night . . . but I don't know how to search for them." (S1)	No search ability; Passive exposure; Accidental discover	Algorithm-driven passive viewing
		"The TV drama just popped up—I don't know how it happened." (S1)	Unable to search; Algorithm dependence	Accidental discovery
		"I don't know how to find the literacy-teaching videos; I just swipe until they appear." (S10)	Accidental learning; Non-intentional access	Unintended learning
		"I watch whatever Kuaishou recommends." (S3)	Algorithm dependence	Reliance on recommendations
	Non-reproducible Skills	"I used to livestream every day, but now I forgot how to start it." (S13)	Forgotten steps; Lost operational path	Unsustainable skills
		"Sometimes I get into the livestream page, but I can't find it again." (S5)	Inability to reproduce operations	Non-replicable action
		"I can't find the singing interface again." (S11)	Broken task path	Fragmented skill memory
Incomplete Affordances	Partial Task Completion	"After buying something, I don't know where the delivery status is—I just wait for the phone call." (S8)	Cannot track logistics	Broken task chain
		"I see messages but can't read them." (S8)	Text barrier	Difficulty interpreting text
		"The wage notice comes, but I can't read it; I must ask my daughter." (S2)	Inability to understand notifications	Text-dependence barrier
	Constrained Self-presentation	"I only send heart emojis—I don't dare type because I might make mistakes." (S10)	Text anxiety; Emoji-only	Restricted expressive ability
		"I can film videos on Douyin but can't choose songs." (S10)	Illiteracy limits personalization	Limited creative capacity
		"I can't interact during livestreams because I can't read." (S13)	Reading difficulty	Social interaction constraints
	Structural Text-dependence Barriers	"I can't read the factory notices—so I wait for someone to explain them." (S2)	Text barrier	Text-centric structural exclusion
		"People ask me if I'm selling today—I don't understand the text." (S9)	Reliance on voice	Text-based exclusion
		"I'm afraid of typing the wrong characters and affecting my business." (S9)	Economic risk	Text-driven inequality

Downloaded from www.emerald.com/itp/article-pdf/39/5/2266/11222077/itp-05-2025-0700en.pdf by University of Guelph user on 04 October 2026

References

Ahmed, S.I., Zaber, M. and Guha, S. (2013), "Usage of the memory of mobile phones by illiterate people", *Proceedings of the 3rd ACM Symposium on Computing for Development*, ACM, Bangalore, India, pp. 1-2, doi: 10.1145/2442882.2442930.

Ahmed, M.A., Islam, M.N., Jannat, F. and Sultana, Z. (2019), "Towards developing a mobile application for illiterate people to reduce digital divide", *2019 International Conference on Computer Communication and Informatics (ICCCI)*, IEEE, Coimbatore, India, pp. 1-5, doi: 10.1109/ICCCI.2019.8822036.

Alfayez, Z.H., Alfayez, B.H. and Abdul-Samad, N.H. (2023), "Read for me: developing a mobile based application for both visually impaired and illiterate users to tackle reading challenge", *I-Com*, Vol. 22 No. 3, pp. 225-240, doi: 10.1515/icom-2023-0031.

Bellard, E. (1999), "The female face of illiteracy in China", *China Information*, Vol. 13 No. 4, pp. 27-65, doi: 10.1177/0920203X9901300402.

Blumer, H. (1954), "What is wrong with social theory?", *American Sociological Review*, Vol. 19 No. 1, pp. 3-10, doi: 10.2307/2088165.

Braun, V. and Clarke, V. (2006), "Using thematic analysis in psychology", *Qualitative Research in Psychology*, Vol. 3 No. 2, pp. 77-101, doi: 10.1191/1478088706qp063oa.

Bucher, T. and Helmond, A. (2018), "The affordances of social media platforms", in Burgess, J., Poell, T. and Marwick, A. (Eds), *The SAGE Handbook of Social Media*, SAGE, London, pp. 233-254.

Chinese Government Network (2021), "Bulletin of the seventh national population census", available at: http://www.gov.cn/guoqing/2021-05/13/content_5606149.htm (accessed 4 October 2024).

Chipchase, J. (2008), "Reducing illiteracy as a barrier to mobile communication", in Katz, J.E. (Ed.), *Handbook of Mobile Communication Studies*, MIT Press, Cambridge, MA, pp. 79-89, doi: 10.7551/mitpress/9780262113120.001.0001.

Correa, T., Pavez, I. and Contreras, J. (2020), "Digital inclusion through mobile phones? A comparison between mobile-only and computer users in internet access, skills and use", *Information, Communication and Society*, Vol. 23 No. 7, pp. 1074-1091, doi: 10.1080/1369118X.2018.1555270.

da Silva, T.H., Furtado, V., Furtado, E., Mendes, M., Almeida, V. and Sales, L. (2024), "How do illiterate people interact with an intelligent voice assistant?", *International Journal of Human-Computer Interaction*, Vol. 40 No. 3, pp. 584-602, doi: 10.1080/10447318.2022.2121219.

Darling-Wolf, F. (2021), "In the city, they go 'pit pit pit': Digital media's affordances and imagined (dis)connections in a rural Japanese community", *New Media and Society*, Vol. 23 No. 7, pp. 1863-1881, doi: 10.1177/1461444820921175.

Doulani, A., Saberi, M.K. and Mirezati, S.Z. (2022), "An investigation of how to retrieve web-based information for illiterate users in Iran", *International Journal of Integrated Supply Management*, Vol. 20 No. 4, pp. 257-274, available at: https://dorl.net/dor/20.1001.1.20088302.2022.20.4.16.5

Fanta-Vagenshtein, Y. (2008), "How illiterate people learn: case study of Ethiopian adults in Israel", *The Journal of Literacy and Technology*, Vol. 9 No. 3, pp. 26-49.

Garg, C. (2021), "Is mobile phone use invading multiple boundaries? A study of rural illiterate women in India", *Indian Journal of Gender Studies*, Vol. 28 No. 1, pp. 29-45, doi: 10.1177/0971521520974845.

Gibson, J.J. (1979), *The Ecological Approach to Visual Perception*, Houghton Mifflin, Boston, MA.

Gou, H. and Gong, X. (2025), "Bridging the digital divide: shopping freedom for illiterate women", *International Conference on Human-Computer Interaction*, Springer Nature Switzerland, Cham, pp. 315-322, doi: 10.1007/978-3-031-94168-9_31.

Guest, G., Bunce, A. and Johnson, L. (2006), "How many interviews are enough? An experiment with data saturation and variability", *Field Methods*, Vol. 18 No. 1, pp. 59-82, doi: 10.1177/1525822X05279903.

Hayat, S., Rextin, A. and Bilal, A. (2021), "Understanding the usability issues in contact management of illiterate and semi-literate users", *PLoS One*, Vol. 16 No. 12, e0259719, doi: 10.1371/journal.pone.0259719.

Downloaded from www.emerald.com/itp/article-pdf/39/5/2266/11222077/itp-05-2025-0700en.pdf by University of Guelph user on 04 October 2026

Hazra, U. and Priyo, A.K.K. (2022), "Unethical practices in online classes during COVID-19 pandemic: an analysis of affordances using routine activity theory", *Journal of Information, Communication and Ethics in Society*, Vol. 20 No. 4, pp. 546-567, doi: 10.1108/JICES-10-2021-0102.

Iqbal, M.W. (2021), "Usability enhancement of SMS interface for illiterate users", *Lahore Garrison University Research Journal of Computer Science and Information Technology*, Vol. 5 No. 3, pp. 31-43, doi: 10.54692/lgurjcsit.2021.0503215.

Irizarry, C., Downing, A. and West, D. (2002), "Promoting modern technology and internet access for under-represented older populations", *Journal of Technology in Human Services*, Vol. 19 No. 4, pp. 13-30, doi: 10.1300/J017v19v04_02.

Islam, M.N., Khan, N.I., Inan, T.T. and Sarker, I.H. (2023), "Designing user interfaces for illiterate and semi-literate users: a systematic review and future research agenda", *Sage Open*, Vol. 13 No. 2, 21582440231172741, doi: 10.1177/21582440231172741.

Jan, S., Maqsood, I., Ahmed, S., Wadud, Z. and Ahmad, I. (2020), "Investigating the use of email application in illiterate and semi-illiterate populations", *Computers, Materials and Continua*, Vol. 62 No. 3, pp. 1473-1486, doi: 10.32604/cmc.2020.08917.

Knoche, H. and Huang, J. (2012), "Text is not the enemy: how illiterates use their mobile phones", *Proceedings of the NUI Workshop*, pp. 1-6.

Lavely, W., Xu, Z., Li, B. and Freedman, R. (1990), "The rise in female education in China: national and regional patterns", *The China Quarterly*, Vol. 121, pp. 61-93, doi: 10.1017/S0305741000013515.

McVeigh-Schultz, J. and Baym, N.K. (2015), "Thinking of you: vernacular affordance in the context of the microsocial relationship app, Couple", *Social Media + Society*, Vol. 1 No. 2, 2056305115604649, doi: 10.1177/2056305115604649.

Medhi, I., Sagar, A. and Toyama, K. (2006), "Text-free user interfaces for illiterate and semi-literate users", *2006 International Conference on Information and Communication Technologies and Development*, IEEE, Bangalore, India, pp. 72-82, doi: 10.1109/ICTD.2006.301841.

Mir, I.U.R. and Wani, M.I. (2025), "Street vendors using smartphones for digital payments: a qualitative study of digital opportunities and inequalities from kashmir region in India", *Journal of Asian and African Studies*, Vol. 60 No. 4, pp. 2490-2504, doi: 10.1177/00219096231218437.

Musiimenta, A., Tumuhimbise, W., Pinkwart, N., Katusiime, J., Mugyenyi, G. and Atukunda, E.C. (2021), "A mobile phone-based multimedia intervention to support maternal health is acceptable and feasible among illiterate pregnant women in Uganda", *Digital Health*, Vol. 7, 2055207620986296, doi: 10.1177/2055207620986296.

Nagy, P. and Neff, G. (2015), "Imagined affordance: reconstructing a keyword for communication theory", *Social Media + Society*, Vol. 1 No. 2, 2056305115603385, doi: 10.1177/2056305115603385.

National Bureau of Statistics (2024), "Illiterate population aged 15 and above by region and sex", available at: https://www.stats.gov.cn/sj/ndsj/2024/indexch.htm (accessed 5 September 2025).

National Bureau of Statistics of China (2021), "Major figures on 2020 population census of China", available at: https://www.stats.gov.cn/sj/pcsj/rkpc/d7c/202111/P020211126523667366751.pdf (accessed 5 September 2025).

Norman, D. (2013), *The Design of Everyday Things: Revised and Expanded Edition*, Basic Books, New York, NY.

Payne, G. (2006), "Re-counting 'illiteracy': literacy skills in the sociology of social inequality", *British Journal of Sociology*, Vol. 57 No. 2, pp. 219-240, doi: 10.1111/j.1468-4446.2006.00107.x.

Rong, X.L. and Shi, T. (2001), "Inequality in Chinese education", *Journal of Contemporary China*, Vol. 10 No. 26, pp. 107-124, doi: 10.1080/10670560124330.

Schrock, A.R. (2015), "Communicative affordances of mobile media: portability, availability, locatability, and multimediality", *International Journal of Communication*, Vol. 9, pp. 1-21.

Downloaded from www.emerald.com/itp/article-pdf/39/5/2266/11222077/itp-05-2025-0700en.pdf by University of Guelph user on 04 October 2026

Stauffer, J., Frost, R. and Rybolt, W. (1978), "Literacy, illiteracy, and learning from television news", *Communication Research*, Vol. 5 No. 2, pp. 221-232, doi: 10.1177/009365027800500205.

Stites, R. (1995), "Limits to the effectiveness of the 1990 anti-illiteracy campaign in promoting literacy among women in a rural Chinese community", *International Journal of Educational Development*, Vol. 15 No. 4, pp. 381-399, doi: 10.1016/0738-0593(94)00051-P.

Thapa, D. and Sein, M.K. (2018), "Trajectory of affordances: insights from a case of telemedicine in Nepal", *Information Systems Journal*, Vol. 285, pp. 796-817, doi: 10.1111/isj.12160.

UNESCO (1978), "Records of the general conference", 20th Session, Paris, October 24 to November 28, 1978: Resolutions (Chinese), UNESCO, Paris.

UNESCO Institute for Statistics (2016), "Literacy", available at: http://uis.unesco.org/en/topic/literacy (accessed 5 April 2025).

World Population Review (2025), "Literacy rate by country 2024", available at: https://worldpopulationreview.com/country-rankings/literacy-rate-by-country (accessed 5 April 2025).

Zhou, M. (2005), "Legislating literacy for linguistic and ethnic minorities in contemporary China", *Current Issues in Language Planning*, Vol. 6 No. 2, pp. 102-121, doi: 10.1080/14664200508668276.

Zhou, Y.Q. and Ding, H. (2020), "The overview and influential factors of digital backfeeding among the three generations of Chinese families", *Chinese Journal of Journalism and Communication*, Vol. 42 No. 3, pp. 6-31, doi: 10.13495/j.cnki.cjjc.2020.03.001.

Further reading

Bouaamri, A. and Barátné Hajdu, Á. (2025), "Literacy and illiteracy in Morocco and the multifaceted role of public libraries", *Library Hi Tech News*, Vol. 42 No. 3, pp. 1-7, doi: 10.1108/LHTN-05-2024-0072.

Fingeret, A. (1983), "Social network: a new perspective on independence and illiterate adults", *Adult Education*, Vol. 33 No. 3, pp. 133-146, doi: 10.1177/074171368303300301.

Lalji, Z. and Good, J. (2008), "Designing new technologies for illiterate populations: a study in mobile phone interface design", *Interacting with Computers*, Vol. 20 No. 6, pp. 574-586, doi: 10.1016/j.intcom.2008.09.002.

Corresponding author
Hao Gao can be contacted at: zhubenhaozi@163.com

For instructions on how to order reprints of this article, please visit our website:
www.emeraldgrouppublishing.com/licensing/reprints.htm
Or contact us for further details: **permissions@emeraldinsight.com**

Downloaded from www.emerald.com/itp/article-pdf/39/5/2266/11222077/itp-05-2025-0700en.pdf by University of Guelph user on 04 October 2026